

Investigating voiced and unvoiced regions of speech for audio deepfake detection

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Abstract—Deep neural network based deepfake detection systems have achieved high levels of accuracy on benchmark datasets and competitions. However, most models lack interpretability. It is challenging to extract reasoning from the network that can convince the human evaluator to trust the decision. Humans often rely on acoustic cues like unnatural pitch jitter, robotic intonation, acoustic artifacts, and unnatural sounding fricatives to judge the quality of the synthetic audio. This study explores the role played by the voiced and unvoiced regions of speech in discriminating synthetic from bonafide speech. A measure of signal periodicity is used to analyze speech into voiced and unvoiced components. Then, the graph attention based AASIST detection system is trained independently on each component. This work compares the accuracy of deepfake detection system using voiced and unvoiced components and analyzes the results on the MLAAD dataset. Our results show that unvoiced regions are particularly more effective in distinguishing synthetic (deepfake) speech from bonafide, and achieves an equal error rate of 6.62%. When combined with voice regions through score-level fusion, the overall performance improves further, yielding a 5.82% EER, a relative improvement of 49% over the baseline system that uses the full audio.

Index Terms—spoofing countermeasure, audio deepfake detection, unvoiced phones

I. INTRODUCTION

The advent of generative AI technologies has revolutionized the speech synthesis, making it possible to generate realistic speech contents that are indistinguishable from human speech [1]. Multi-speaker speech synthesis systems now clone anyone’s voice using just a few seconds of enrollment audio [2]. These technologies combined with the ubiquity of everyone’s voice on various open internet based platforms makes everyone vulnerable to voice spoofing attacks. This not only threatens personal security but also undermines the integrity of automatic speaker verification (ASV) system, making it crucial to develop robust spoofing countermeasure for secure ASV. The ability to detecting synthetic speech across various communication platforms is becoming really important to maintain the trust in voice communication over digital channels.

Synthetic speech detection has been an active area of research for over a decade [3], [4]. The goal of this task to distinguish bonafide speech samples from text-to-speech and voice conversion samples. Ongoing research into novel neural network architectures and loss functions has significantly

advanced the state-of-the-art across a wide range of synthetic systems [4]–[6], including in-the-wild conditions [7]. The current state-of-the-art systems process raw speech waveforms as input, feeding them through a deep convolutional neural network, followed by temporal pooling of features to predict whether the presented speech utterance is bonafide or synthetic [8]–[11].

Explainable models that support their predictions with specific evidence from the input space can help build human trust in these black-box systems and may even assist forensic experts in identifying subtle signatures of synthetic speech. Extracting interpretable explanations for a DNN model’s predictions is a challenging task and is relatively a new area of research. Authors in [12] studied several explainable artificial intelligence (XAI) methods used for image classification. Particularly, they looked at Deep Taylor [13], integrated gradients [14], and layer-wise relevance propagation [15]. Their empirical results concluded that those approaches tend to focus on the regions of unvoiced speech as well as the first three formants when voiced speech. Similarly, a recent study [16] that applied SHapley Adaptive exPlanations (SHAP) [17] to identify target regions in the speech spectrum found artefacts associated to particular regions in the voiced and unvoiced speech frames.

Speech production involves a complex coordination of several muscular movements. The vibration of the vocal cords play a central role in producing vowel sounds which form the major content of speech [18]. But the vocal fold vibrations are intermittently stopped to produce unvoiced sounds like unvoiced stop consonants and fricatives. Free flowing speech involves a rhythmic dance between the voiced and unvoiced states of vocal folds. Unvoiced fricative signals look similar to white noise but are often colored by the adjacent sounds due to co-articulation. The extent of such co-articulatory effects in unvoiced sounds is not uniform in bonafide speech. In contrast, text-to-speech synthesis and voice conversion systems often produce unnatural waveforms for the fricatives and stops. Hence the unvoiced regions could hold important cues for discriminating bonafide from synthetic speech.

In the literature, phoneme specific models have been explored for detecting replay attacks with fair success. [19]

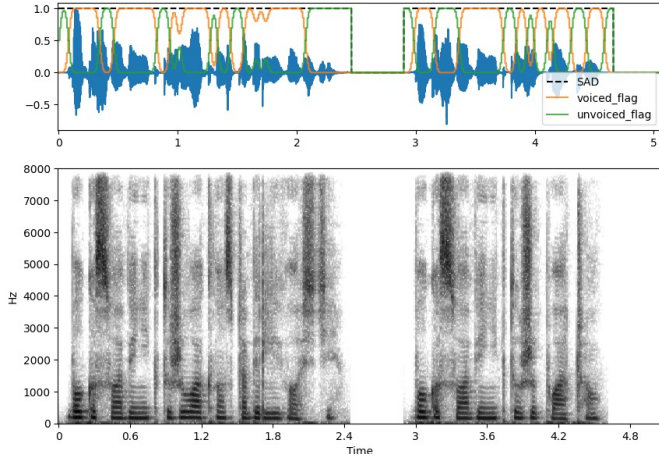


Fig. 1. Spectrogram of an example utterance from the MLAAD train set. Top panel shows the waveform along with the speech activity detection (SAD), voiced and unvoiced smoothed flags. The bottom panel shows the spectrogram of the utterance, highlighting its acoustic features.

studied the impact of the phoneme-based pop-noise of the microphone, while [20] directly explored the use of voiced and unvoiced segments. They both achieve high accuracy on the ASVspoof 2017 dataset [21]. Additionally, for synthetic speech detection, studies such as [22]–[25] highlighted the importance of silence regions for spoofing detection. Most of their work rely on the ASVspoof 2019 dataset [26] that has silence at the beginning and end of the speech utterances, making it difficult to judge the significance of voice and unvoiced segments.

However to the best of our knowledge, we are the first to explore the use of voiced and unvoiced phones to detect synthetic speech. We conducted our experiments using the recently proposed multi-lingual audio anti-spoofing (MLAAD) dataset [27]. Fig 1 illustrates an audio example from the MLAAD dataset, with both the waveform and the spectrogram. It also shows the voiced and unvoiced regions. Our goal is to address the following research questions:

- Can existing spoofing detection systems accurately detect synthetic speech by analyzing separately the voiced and unvoiced regions?
- Are the unvoiced or voiced segments complementary in improving detection performance?

The remainder of this paper is organized as follows. Section II describes our methodology for segmenting speech utterances into voice and unvoiced components, and model architecture details. Our experiments and results are presented in Section IV followed by conclusion in Section V.

II. VOICED AND UNVOICED-BASED SYNTHETIC SPEECH DETECTION

In this section, we explain the method used to analyze the speech signals by dividing them into voiced and unvoiced

components. We also describe the model architecture used for the deepfake detection.

A. Voiced and unvoiced segmentation

Speech sounds can be categorized into two fundamental components: voiced and unvoiced phones. Voiced sounds occur when the vocal folds vibrate during the speech production, whereas unvoiced sounds do not. In the literature several algorithms have been proposed to decompose speech signal into these components [28]–[31]. This decomposition has been applied for various purposes such as voice analysis, acoustic phonetics and speech recognition. In our work, we focus on the temporal segmentation of speech into voiced and unvoiced regions using the pitch detection and estimation that can be used as a measure of voicing. There are several existing algorithms to detect and estimate the pitch [32]–[35]. Most of them are based on peak finding on the auto-correlation function. In this paper, we use the probabilistic YIN (pYIN) algorithm to identify whether a given speech frame is voiced, using the implementation from the Librosa library [36]. We analyze the speech signal with a frame-length of 80 ms and a hop-length of 10 ms to estimate the pitch, voicing probability and voicing flag. The binary voicing-flag estimated by pYIN¹ is used to segment the speech signal. This approach is particularly effective when working with clean datasets where non-speech noise is minimal.

To improve the segmentation process, we smoothed the voiced and unvoiced flags using a Hamming window with a width of 10 frames. This smoothing helps to reduce the abrupt bursts in the speech signal due to discontinuities. In order to exclude the non-speech silences from our analysis, we used the Web-RTC based speech activity detection (SAD) implementation² with a severity parameter of 2 to get the speech/non-speech binary mask. The input audio signal x is multiplied with the binary SAD mask to obtain the x_{speech} signal.

$$x_{speech} = SAD(x) * x, \quad (1)$$

The x_{speech} signal is multiplied with the smoothed voicing flag to generate the voiced component (x_{voi}), and is multiplied with the unvoiced flag to generate the unvoiced speech (x_{unv}) as follows:

$$x_{voi} = voicing_flag(x) * x_{speech}, \quad (2)$$

$$x_{unv} = (1 - voicing_flag(x)) * x_{speech}, \quad (3)$$

Fig 1 shows the SAD , $voiced_flag$, and $unvoiced_flag$ for an example utterance from the MLAAD training subset. Note that after smoothing, the voiced and unvoiced flags may overlap slightly at the boundaries, which helps include the transition regions and also minimizes the discontinuities in the speech signal. In this work, we performed deepfake detection experiments using the full audio x , SAD segmented x_{speech} , voiced component x_{voi} , and unvoiced component x_{unv} . The

¹<https://librosa.org/doc/main/generated/librosa.pyin.html>

²<https://github.com/wiseman/py-webrtcvad>

TABLE I
MLAAD TRAIN, DEVELOPMENT AND EVALUATION PROTOCOLS FOR
BINARY CLASSIFICATION TASK.

Types	Partitions		
	train	dev	eval
Bonafide	33225	3656	4438
Synthetic	54867	6133	5000

same model architecture was used in all four scenarios. This architecture is described in the next section.

B. Synthetic detection model

In the literature several Deep Neural Networks (DNNs) have been proposed to detect the synthetic speech. Recent research has extensively explored large speech foundation models like Wav2vec2.0 [37], Conformer [38], WavLM [39] and Whisper [40] to achieve the state-of-the-art performance. The main focus of our work is to highlight the importance of voiced and unvoiced regions in the speech for explainable audio deepfake detection, rather than achieving state-of-the-art performance. Hence, we selected the widely-used AA-SIST [10] model. ASSIST is a graph attention based system that employs a RawNet2-based encoder [8] to extract higher-level spectro-temporal features representation directly from the raw waveform. RawNet2 encoder consists of a bank of 70 sinc filters and six residual blocks to extract the spectral and temporal representations. A heterogeneous graph attention layers and max graph operations are then used to model the local spectro-temporal dependency. Finally, the output scores are generated using a readout operation and a linear output layer.

III. DATASETS

We performed our experiments using the MLaAD dataset [27]. It consists of 52 different state-of-the-art synthetic speech generation systems [27]. More details about the generation systems can be found in the Coqui-TTS [41] and HuggingFace repositories. We manually labelled the acoustic models and vocoders based on the available metadata. For experiments, we divide the data into three disjoint partitions: train, development, and evaluation sets. Bonafide samples were sourced from the multilingual M-AILABS dataset [42]. The training and development partitions were created with a set of 9 different attacks (vits, xtts-v1, xtts-v2, tacotron2, tacotron2-dca, tacotron2-ddc, glow-tts and neural-hmm), whereas the evaluation set was created with a set of 5 unseen attacks (bark, capacitrn, fastpitch, overflow and tortoise-tts). Details of the data partitions are presented in Table I.

IV. EXPERIMENTS AND RESULTS

The objective of the experiments in this paper is to assess the importance of voiced and unvoiced components of speech for deepfake detection. We trained four ASSIST models on each of the following input types - *full* audio, SAD segmented *speech*, *voiced*, and *unvoiced* components. The model architecture was trained on each of these conditions with segments of approximately 4 seconds duration (64,600 samples). We

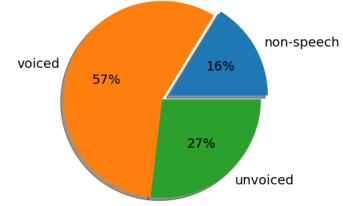


Fig. 2. Proportion of utterance duration attributed to non-speech, voiced, and unvoiced regions in the train subset of MLaAD.

followed the same training recipe and model configurations available in this repository³. For training, we used the Adam optimiser with a mini-batch size of 16, a fixed learning rate of 0.0001, and trained for 30 epochs. The best model was selected based on the lowest equal error rate (EER) on the development set.

A. Standalone results

We first assess independently the performance of all four systems on the evaluation set of MLaAD as detailed in Table II. The baseline system trained on full utterances achieved an EER of 11.40%. The speech-only system achieved slightly better results with an EER of 10.10%. Additionally, the voiced-only system got slightly worse EER of 12.26%. But interestingly, the unvoiced-only system achieved an EER of 6.62%, much lower than the three previous scenarios. A detailed breakdown of these results across attack types is also provided in columns 3-7 of Table II. It shows the false acceptance rate (FAR) of each attack type at the threshold corresponding to the EER on the full evaluation set. Based on this analysis, it is interesting to see that the unvoiced-only system works very well on the synthetic speech generated by Bark, capacitrn, fastpitch and overflow, but it does not perform well on tortoise-tts samples. This is possibly because its mean spectrum is much closer to bonafide than the other synthetic systems.

B. Fusion results

We use the linear logistic regression approach from scikit-learn to combine the scores from both voiced and unvoiced systems. We applied mean and std normalization on both voice and unvoiced development scores and trained logistic regression fusion model on them. The fusion results are shown in the last row of Table II. The EER of the combined system is 5.82%, which shows significant improvement of 52% over the voiced system and 12% over the unvoiced system. Our fusion results clearly show that both voice and unvoiced systems are complementary in improving deepfake detection performance. More importantly, the combined system yields 49% reduction in EER (from 11.40% to 5.82%) compared to the standard technique that uses the full audio.

C. Analysis

The duration of an utterance consists of both non-speech and speech regions. The speech region in turn consists of

³<https://github.com/clovaai/aasist/tree/main>

TABLE II
POOLED EER AND ATTACK-WISE RESULTS IN TERMS OF FAR FOR THE MLAAD EVALUATION SET ON DIFFERENT SCENARIOS.

Scenario	Pooled EER	FAR of attacks @ EER threshold				
		bark	capacitron	fastpitch	overflow	tortoise tts
full-audio	11.40%	10.90%	6.30%	0.00%	0.00%	39.80%
speech-only	10.10%	15.70%	16.20%	0.00%	0.00%	18.60%
voiced	12.26%	30.10%	14.20%	0.10%	0.00%	16.90%
unvoiced	6.62%	2.10%	1.60%	0.10%	0.00%	29.30%
voiced+unvoiced	5.82%	7.80%	3.10%	0.10%	0.00%	18.10%

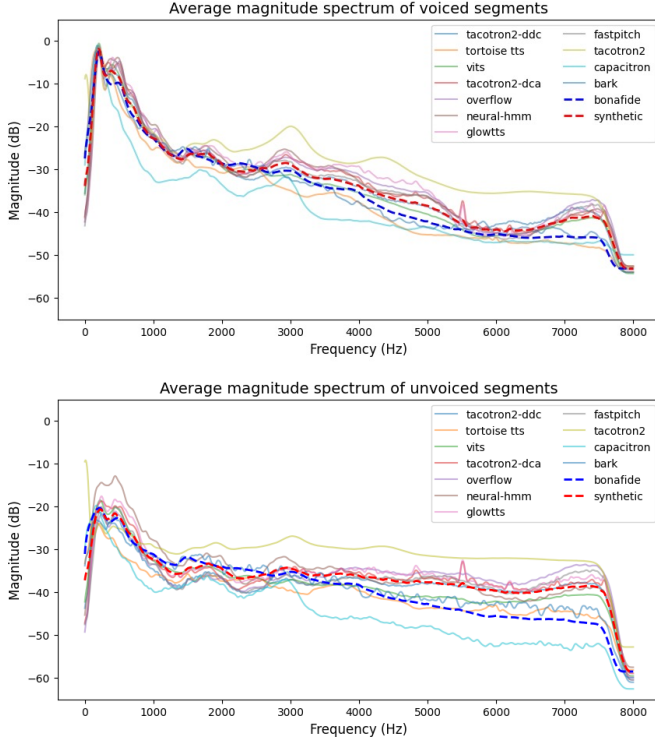


Fig. 3. Average spectrum of the voiced and unvoiced segments for bonafide and synthetic utterances from the MLAAD dataset.

both voiced and unvoiced regions. Fig 2 shows the pie chart of the proportion of an utterance attributed to non-speech, voiced and unvoiced regions. The duration of the unvoiced segments of speech are on an average only 27% of the audio duration. Yet, the unvoiced components outperform the voiced components for deepfake detection. We investigated the spectrum of the signal to understand why the unvoiced regions of speech contain more discriminative features for deepfake detection. Fig 3 shows the average spectrum of bonafide and synthetic classes. We see that the average spectra of bonafide and synthetic utterances are more overlapping for the voiced segments compared to that of unvoiced segments. For the unvoiced segments, there is a clear departure of the synthetic speech spectrum in the higher frequencies. Fig 4 illustrates the spectral mean and standard deviations for Capacitron and Tortoise tts spoofing attack types. We observe that the spectra of the unvoiced segments are well separated in the higher frequencies for the Capacitron TTS spoofing attack. However,

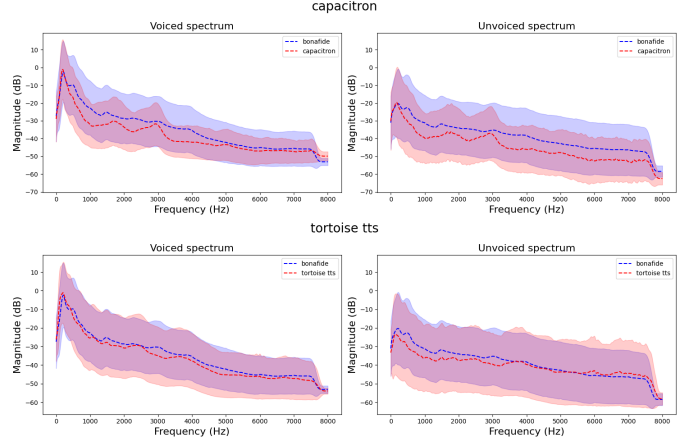


Fig. 4. Comparing the spectral mean and standard deviation of bonafide, fastpitch and tortoise-tts samples.

the Tortoise-TTS model produces a spectrum much closer to bonafide speech which makes it harder to detect compared to the other attack types in the evaluation set. Perhaps the TTS systems which are optimized for perceptual quality, tend to synthesize the lower frequency spectrum better than the higher frequencies. The higher frequency spectral artefacts in the synthetic speech might go unnoticed to the human auditory system. We also observed that the FAR of the model on fastpitch, and overflow TTS was consistently close to 0% across all the systems. The fastpitch and overflow models used in creating the MLAAD dataset were both single speaker TTS models trained on the LJSpeech dataset. That might be the reason for the 100% accuracy on these two attack types.

V. CONCLUSIONS

In this paper we studied the role of voiced and unvoiced speech regions for synthetic speech detection. Our experimental results, performed on the MLAAD dataset, show that the deepfake detection system trained on the unvoiced regions achieve better EERs compared to the systems trained on the full audio, speech-only regions or voiced regions. Fusion experiments also show that voice and unvoiced systems are complementary. The combined system yields 49% relative reduction in EER over the baseline system that uses the full audio. Future work will investigate an attention-based single end-to-end system that can possibly learn key features from both voiced and unvoiced regions.

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